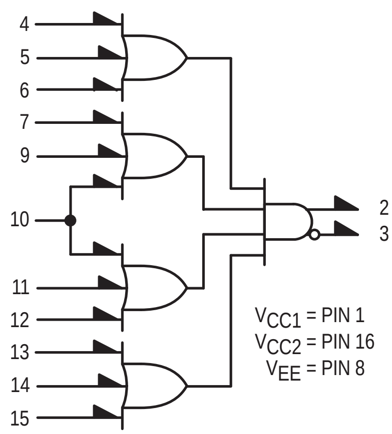
MC10121

4-Wide OR-AND/$\overline{\text{OR-AND}}$ Gate

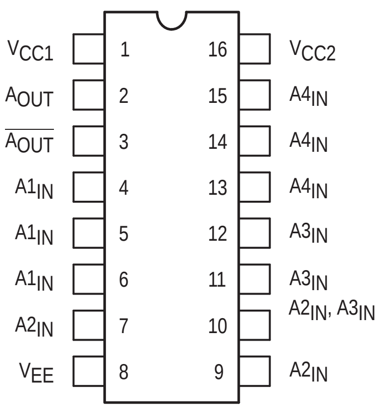
The MC10121 is a basic logic building block providing the simultaneous OR–AND/OR–AND–Invert function, useful in data control and digital multiplexing applications.

- P_D = 100 mW typ/pkg (No Load)
- t_{pd} = 2.3 ns typ
- t_r, t_f = 2.5 ns typ (20%–80%)

LOGIC DIAGRAM



DIP
PIN ASSIGNMENT

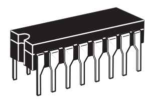


Pin assignment is for Dual–in–Line Package.
For PLCC pin assignment, see the Pin Conversion Tables on page 18.

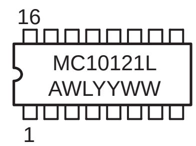
ON Semiconductor

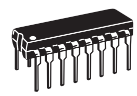
http://onsemi.com

MARKING DIAGRAMS

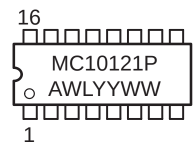


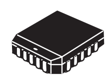
CDIP–16
L SUFFIX
CASE 620



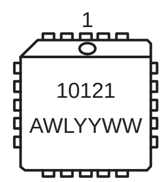


PDIP–16
P SUFFIX
CASE 648





PLCC–20
FN SUFFIX
CASE 775



A = Assembly Location
WL = Wafer Lot
YY = Year
WW = Work Week

ORDERING INFORMATION

Device	Package	Shipping
MC10121L	CDIP–16	25 Units / Rail
MC10121P	PDIP–16	25 Units / Rail
MC10121FN	PLCC–20	46 Units / Rail

ELECTRICAL CHARACTERISTICS

Characteristic	Symbol	Pin Under Test	Test Limits –30°C Min	–30°C Max	+25°C Min	+25°C Typ	+25°C Max	+85°C Min	+85°C Max	Unit
Power Supply Drain Current	I_E	8		29		20	26		29	mAdc
Input Current	I_{inH}	7		390			245		245	µAdc
		9		390			245		245	
		10		495			310		310	
	I_{inL}	7	0.5		0.5			0.3		µAdc
		9	0.5		0.5			0.3		
		10	0.5		0.5			0.3		
Output Voltage Logic 1	V_{OH}	3	–1.060	–0.890	–0.960		–0.810	–0.890	–0.700	Vdc
		2	–1.060	–0.890	–0.960		–0.810	–0.890	–0.700	
Output Voltage Logic 0	V_{OL}	3	–1.890	–1.675	–1.850		–1.650	–1.825	–1.615	Vdc
		2	–1.890	–1.675	–1.850		–1.650	–1.825	–1.615	
Threshold Voltage Logic 1	V_{OHA}	3	–1.080		–0.980			–0.910		Vdc
		2	–1.080		–0.980			–0.910		
Threshold Voltage Logic 0	V_{OLA}	3		–1.655			–1.630		–1.595	Vdc
		2		–1.655			–1.630		–1.595	
Switching Times (50Ω Load)										ns
Propagation Delay	t_{4+3-}	3	1.4	3.6	1.4	2.3	3.4	1.4	3.5	
	t_{4-3+}	3	1.4	3.6	1.4	2.3	3.4	1.4	3.5	
	t_{4+2+}	2	1.4	3.6	1.4	2.3	3.4	1.4	3.5	
	t_{4-2-}	2	1.4	3.6	1.4	2.3	3.4	1.4	3.5	
Rise Time (20 to 80%)	t_{3+}	3	0.9	4.1	1.1	2.5	4.0	1.1	4.6	
	t_{2+}	2	0.9	4.1	1.1	2.5	4.0	1.1	4.6	
Fall Time (20 to 80%)	t_{3-}	3	0.9	4.1	1.1	2.5	4.0	1.1	4.6	
	t_{2-}	2	0.9	4.1	1.1	2.5	4.0	1.1	4.6	

ELECTRICAL CHARACTERISTICS (continued)

			TEST VOLTAGE VALUES (Volts)					
@ Test Temperature			**V_{IHmax}**	**V_{ILmin}**	**V_{IHAmin}**	**V_{ILAmax}**	**V_{EE}**	
–30°C			–0.890	–1.890	–1.205	–1.500	–5.2	
+25°C			–0.810	–1.850	–1.105	–1.475	–5.2	
+85°C			–0.700	–1.825	–1.035	–1.440	–5.2	
			TEST VOLTAGE APPLIED TO PINS LISTED BELOW					
Characteristic	**Symbol**	**Pin Under Test**	**V_{IHmax}**	**V_{ILmin}**	**V_{IHAmin}**	**V_{ILAmax}**	**V_{EE}**	**(V_{CC}) Gnd**
Power Supply Drain Current	I_E	8					8	1, 16
Input Current	I_{inH}	7 9 10	7 9 10				8 8 8	1, 16 1, 16 1, 16
	I_{inL}	7 9 10		7 9 10			8 8 8	1, 16 1, 16 1, 16
Output Voltage Logic 1	V_{OH}	3 2	 4, 10, 13				8 8	1, 16 1, 16
Output Voltage Logic 0	V_{OL}	3 2	4, 10, 13 				8 8	1, 16 1, 16
Threshold Voltage Logic 1	V_{OHA}	3 2	 10, 13		 4	4 	8 8	1, 16 1, 16
Threshold Voltage Logic 0	V_{OLA}	3 2	 10, 13		4 	 4	8 8	1, 16 1, 16
Switching Times (50Ω Load)			**+1.11V**		**Pulse In**	**Pulse Out**	**–3.2 V**	**+2.0 V**
Propagation Delay	t_{4+3-} t_{4-3+} t_{4+2+} t_{4-2-}	3 3 2 2	10, 13 10, 13 10, 13 10, 13		4 4 4 4	3 3 2 2	8 8 8 8	1, 16 1, 16 1, 16 1, 16
Rise Time (20 to 80%)	t_{3+} t_{2+}	3 2	10, 13 10, 13		4 4	3 2	8 8	1, 16 1, 16
Fall Time (20 to 80%)	t_{3-} t_{2-}	3 2	10, 13 10, 13		4 4	3 2	8 8	1, 16 1, 16

Each MECL 10,000 series circuit has been designed to meet the dc specifications shown in the test table, after thermal equilibrium has been established. The circuit is in a test socket or mounted on a printed circuit board and transverse air flow greater than 500 linear fpm is maintained. Outputs are terminated through a 50-ohm resistor to –2.0 volts. Test procedures are shown for only one gate. The other gates are tested in the same manner.